

Multimodal data integration study

■ To do ■ In progress ■ Done ■ Not concerned

PROJECT SETUP

Human Resources

Project manager
Clinician/Biologist
Biostatistician/Bioinformatician/Computational Biologist
Data steward
Ethicist/Legal expert



Computer Resources

HPC/GPU access
Short-term storage
Long-term storage
Energy footprint assessed



Good practices

FAIR principles
CARE principles
Open code/data policy defined
Licences



INTEGRATION PROTOCOL

RESEARCH QUESTION

Formulation

Precise biological/clinical question
Outcomes and measures specified
Success criteria defined



DATA PREPARATION

Governance

Regulatory compliance and re-identification risk
Access, authorisations and consent
Metadata documented
Data Management Plan



ALGORITHM SELECTION

Task formulation

Task type defined
Learning strategy chosen
Algorithm assumptions checked against data



Scope

Population/system defined
Rationale for multimodality
Exploratory or inferential objectives
Interpretability requirement
Sample size/Power/Biological replicates



Per-modality preparation

Modalities to collect and/or reuse defined
QC and preprocessing documented
Confounders characterised (batch, bias ...)
Exploratory analysis



Multimodal integration

Integration timing (early/intermediate/late)
Intermediate fusion approach (if applicable)
Prior resources incorporated



Cross-modal connections

Connection type (vertical/horizontal ...)
Missing modality pattern
Redundancy/complementarity
Data harmonisation



Practical constraints

Interpretability approach defined
Computational feasibility
Software availability and maintenance



EVALUATION

Validation

Internal validation strategy
External validation and representativeness
Data leakage risks controlled



Metrics

Metrics suited to task
Unimodal baseline established
Clinical utility considered



Multimodal evaluation

Modality ablation and dropout
Cross-modal validation
Modality importance



Reproducibility

Version control (data, script ...)
Workflow manager
Documentation
Benchmark/data challenge
Pre-registration

